Supplementary: Evaluating Contextual Bandit-Based Recommendation for Mental Health Interventions in Real-World Student Settings

1 Appendix

1.1 Prior Estimates of Interventions Across Activity Contexts

Table A.1. Prior probabilities of different interventions across various activity contexts. Probabilities marked with an asterisk * are below 0.4 and were excluded from implementation within that context. Acronyms - PA: Physical activity, MR: Mental relaxation, CA: Cognitive activities, ESE: Emotional and social engagement.

Intervention/ Context	Attending Lecture	Exercise	Relaxing	In vehicle	Cycling	Walking	Running	Studying	Eating	Standing
Breathing Exercise **(PA)**	0.842	0.842	0.947	0.526	0.368*	0.789	0.474	0.842	0.158*	0.972
Calling or texting loved ones **(ESE)**	0.211*	0.579	0.972	0.972	0.368*	0.947	0.316*	0.579	0.895	0.972
Close your eyes and try to hear ambient sounds **(MR)**	0.526	0.789	0.972	0.789	0.474	0.789	0.421	0.895	0.526	0.972
Doing Simple Neck Rolls to Release Tension **(PA)**	0.895	0.972	0.972	0.895	0.263*	0.895	0.316*	0.92	0.421	0.972
Eating something you like **(ESE)**	0.158*	0*	0.972	0.632	0*	0.474	0*	0.842	0.972	0.789
Go for a small walk **(PA)**	0.053*	0.579	0.737	0*	0*	0.789	0.368*	0.895	0.105*	0.789
Journal Writing **(CA)**	0.368*	0*	0.789	0.158*	0*	0.053*	0.053*	0.632	0.053*	0.421
Listening to Music **(MR)**	0.211*	0.972	0.972	0.947	0.895	0.972	0.867	0.789	0.842	0.972
Make a list of positive things inside you **(CA)**	0.684	0.421	0.947	0.474	0.211*	0.368*	0.211*	0.789	0.368*	0.842
Observe your surroundings **(ESE)**	0.842	0.947	0.972	0.972	0.842	0.972	0.842	0.895	0.947	0.972
Play mobile game **(CA)**	0.263*	0.105*	0.842	0.474	0*	0.263*	0.053*	0.368*	0.263*	0.737
Scroll through old memories from your gallery **(ESE)**	0.421	0.316*	0.947	0.842	0.158*	0.684	0.158*	0.526	0.632	0.895
Stretching **(PA)**	0.368*	0.947	0.972	0.158*	0.053*	0.737	0.368*	0.895	0.105*	0.947
Watching funny videos **(MR)**	0.158*	0.053*	0.972	0.842	0.211*	0.789	0*	0.368*	0.842	0.842
Watching motivational video **(MR)**	0.158*	0.579	0.842	0.684	0.211*	0.579	0.105*	0.526	0.632	0.789
Writing Down a Worry and Putting It Aside **(CA)**	0.632	0.211*	0.895	0.368*	0*	0.158*	0*	0.789	0.158*	0.684

1.2 Distribution of Intervention Responses

Table A.2. Frequency distribution of "Yes," "Yes but not feasible," and "No" across 10 self-reported activity contexts

Context	Yes	Yes, but not feasible	No	Total
Attending Lecture	402	112	198	712
Cycling	22	5	11	38
In vehicle	38	9	17	64
Eating	404	69	120	593
Exercise	131	14	30	175
Relaxing	1661	325	1164	3150
Running	41	5	27	73
Standing	257	63	137	457
Studying	1066	253	554	1873
Walking	699	84	244	1027
Total	4721	939	2502	8162

Table A.3. Frequency distribution of "Yes," "Yes but not feasible," and "No" across self-reported social contexts

Context	**Yes**	**Yes, but not feasible**	**No**	**Total**
Alone	2985	467	1575	5027
With Someone (Engaged in Conversation)	1114	314	560	1988
With Someone (Not Engaged in Conversation)	622	158	367	1147
Total	4721	939	2502	8162

1.3 Posterior Estimates of Interventions Across Activity Contexts - Post Deployment

Table A.4. Posterior probabilities (with confidence interval) of interventions across self-reported activity contexts. Acronyms - Int: Interventions, PA: Physical activity, MR: Mental relaxation, CA: Cognitive activities, ESE: Emotional and social engagement. Cells marked with "–" denote intervention–context combinations that were either not deployed due to low prior feasibility (Table A.1) or insufficiently observed during deployment.

Int/ Context	Attending Lecture	Exercise	Relaxing	In vehicle	Cycling	Walking	Running	Studying	Eating	Standing
I1 (PA)	0.47 [0.40-0.55]	-	0.56 [0.50-0.62]	-	-	0.63 [0.44-0.80]	-	0.61 [0.53-0.68]	-	0.56 [0.46-0.67]
I2 (ESE)	-	-	0.48 [0.43-0.54]	0.38 [0.19-0.59]	-	0.44 [0.37-0.51]	-	-	0.35 [0.25-0.46]	0.41 [0.31-0.51]
I3 (MR)	-	-	0.62 [0.57-0.66]	-	-	0.42 [0.17-0.69]	-	0.56 [0.50-0.61]	-	0.56 [0.46-0.66]
I4 (PA)	0.56 [0.51-0.61]	0.77 [0.67-0.85]	0.61 [0.55-0.66]	-	-	0.59 [0.49-0.69]	-	0.76 [0.72-0.79]	-	0.61 [0.50-0.71]
I5 (ESE)	-	-	0.43 [0.37-0.49]	-	-	-	-	0.35 [0.27-0.43]	0.74 [0.69-0.79]	-
I6 (PA)	-	-	-	-	-	0.62 [0.38-0.84]	-	0.22 [0.17-0.28]	-	-
I7 (CA)	-	-	0.16 [0.04-0.35]	-	-	-	-	-	-	-
I8 (MR)	-	0.63 [0.50-0.75]	0.49 [0.44-0.55]	0.47 [0.23-0.71]	0.54 [0.35-0.71]	0.55 [0.49-0.60]	0.51 [0.37-0.64]	0.38 [0.27-0.49]	0.26 [0.13-0.40]	0.47 [0.35-0.59]
I9 (CA)	0.30 [0.13-0.51]	-	0.36 [0.30-0.43]	-	-		-	0.19 [0.10-0.31]	-	-
I10 (ESE)	0.67 [0.61-0.73]	0.77 [0.62-0.89]	0.76 [0.73-0.80]	0.84 [0.70-0.95]	0.76 [0.56-0.91]	0.88 [0.85-0.91]	0.69 [0.54-0.82]	0.78 [0.74-0.81]	0.82 [0.77-0.87]	0.82 [0.74-0.88]
I11 (CA)	-	-	0.34 [0.21-0.48]	-	-	-	-	-	-	-
I12 (ESE)	-	-	0.40 [0.33-0.46]	-	-	-	-	-	-	0.36 [0.14-0.61]
I13 (PA)	-	0.82 [0.68-0.93]	0.52 [0.46-0.57]	-	-	-	-	0.41 [0.35-0.47]	-	0.42 [0.28-0.58]
I14 (MR)	-	-	0.48 [0.43-0.54]	-	-	0.12 [0.02-0.30]	-	-	0.26 [0.13-0.41]	-
I15 (MR)	-	-	0.21 [0.10-0.35]	-	-	-	-	-	-	-
I16 (CA)	0.18 [0.04-0.38]	-	0.24 [0.16-0.34]	-	-	-	-	0.12 [0.05-0.23]	-	-

*Note: **I1** - Breathing Exercise, **I2** - Calling or texting loved ones, **I3** - Close your eyes and try to hear ambient sounds, **I4** - Doing Simple Neck Rolls to Release Tension, **I5** - Eating something you like, **I6** - Go for a small walk, **I7** - Journal Writing, **I8** - Listening to Music, **I9** - Make a list of positive things inside you, **I10** - Observe your surroundings, **I11** - Play mobile game, **I12** - Scroll through old memories from your gallery, **I13** - Stretching, **I14** - Watching funny videos, **I15** - Watching motivational video, **I16** - Writing Down a Worry and Putting It Aside.*